

Workshop 5

Monday tutorial? Pick another one to attend this week

Welcome!

Find a group to work with & say hi!

ECO 101: Principles of Microeconomics

Workshop Overview

- Measuring value created by a market
- Individuals to markets
- Who gets the surplus when things change?

Measuring value created by a market

Revisiting W1's Benefit vs Opp. Cost comparison

W1 Concepts you already have

- **Consumer surplus:** what you were willing to pay for it minus what you paid.
 - W1's Benefit vs Opportunity Cost, rearranged to get $(M)WTP$ vs Price. This week: $MWTP - P$
- **Producer surplus:** what you were paid for the good minus Marg. Cost to make (or the min. you're willing to accept to sell it)
 - W1's P vs WTA or MC. This week: $P - MC$
- **Total (Societal) surplus** = Sum of surplus for everyone participating or affected by the trades
 - If only consumers and producers: $CS + PS$

Efficiency

Market Allocation: Quantity traded and who bought/produced the goods traded

A market or any allocation is **efficient** when

1. For the goods already being traded, no **reshuffling** of who holds/produced could make someone better off without making someone else worse off
2. We are trading all good for which the benefit to **society** is at least as high as the social cost of producing it, and none for which it is lower.

Notice: W2 concepts of gains from trade. Does not consider fairness or prices

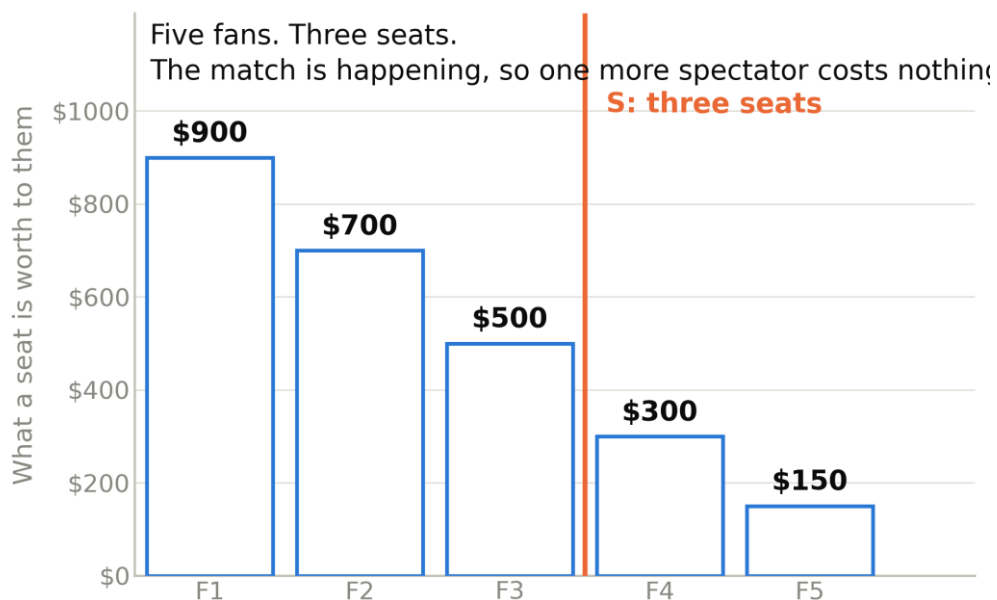
Total (Market) Surplus

Societal Surplus

- Qth unit: $\text{Surplus} = MB_{\text{society}} - MC_{\text{society}}$
- Total: Total Social Benefit – Total Social Cost
- In perfectly competitive markets (no externalities)
 - $MB_{\text{society}} = MWTP$
 - $MC_{\text{society}} = MC$
- $\text{Surplus} = MWTP - MC$
- $= MWTP - p + p - MC$

Five fans, three seats

- Only three stadium seats. No more capacity possible
- Stadium is built, match happening, major costs are **sunk**. The marginal cost of letting one more person in is effectively zero.
- Tickets priced at \$100 each



Five fans, three seats

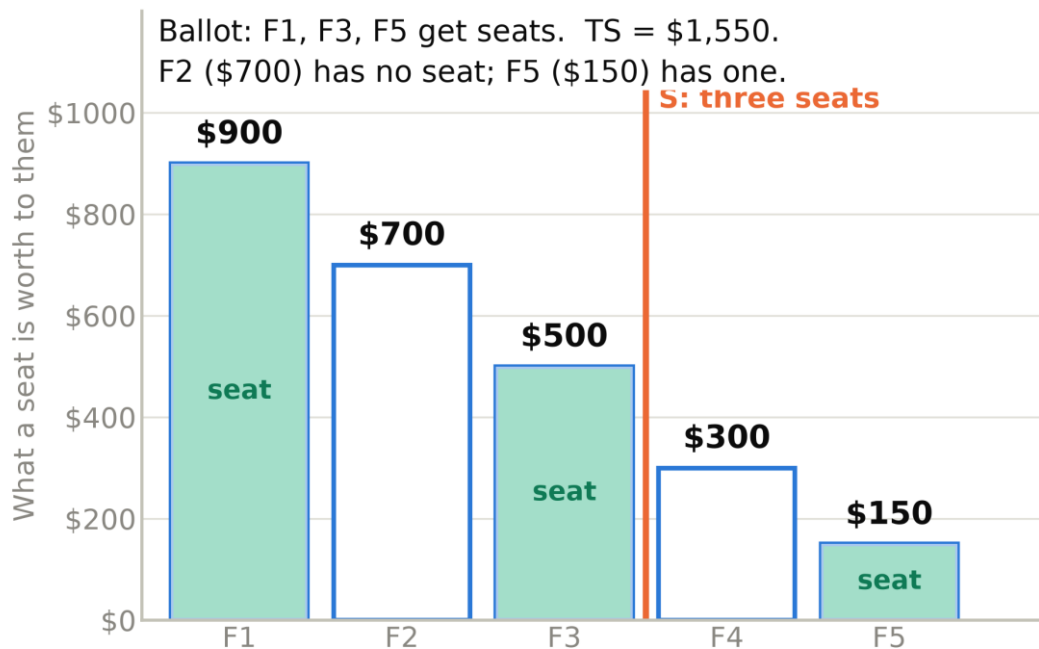
- Who gets the tickets? Each buys only 1 if they buy:

Fan	F1	F2	F3	F4	F5
WTP	\$900	\$700	\$500	\$300	\$150

- Total Surplus:
- Consumer Surplus:
- Producer Surplus:
- Efficient allocation:

Same quantity, wrong people

- FIFA runs a lottery instead. F1, F3 and F5 win.
- Three seats filled, as in the efficient allocation, but $TS = \$1,550$, not $\$2,100$.
 - Lost: $\$550 = F2\text{'s } \$700 - F5\text{'s } \$150$.

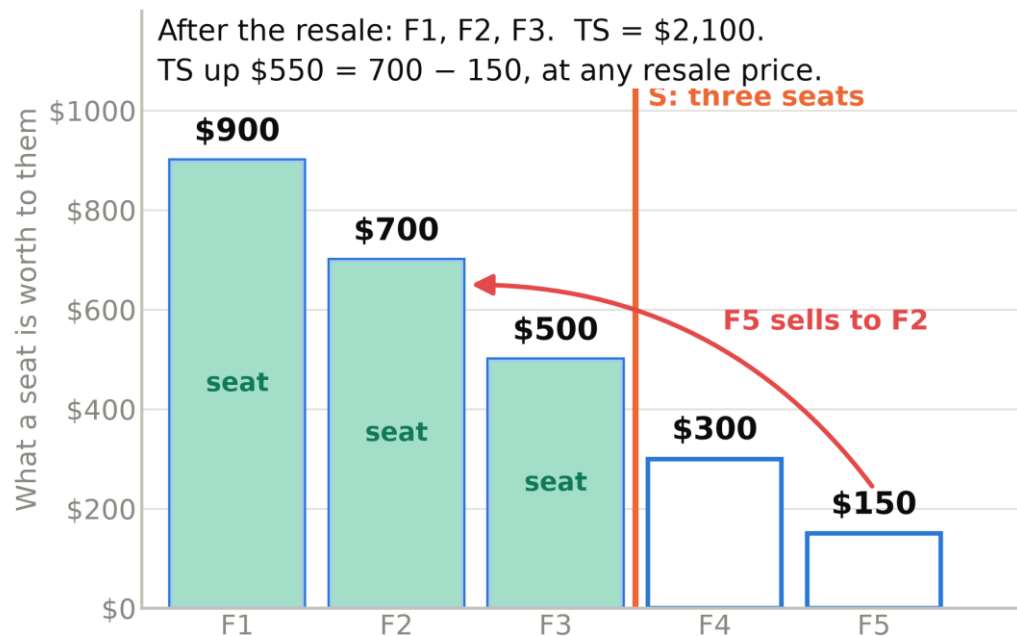


Worksheet Q1 (counts for grades)

Put answer on worksheet

Trade fixes misallocation

- F5 (\$150) sells to F2 (\$700) at any price in between: both better off. TS up \$550; the price enables trade
- Trade continues until no one without a seat values it more than someone with one. **Efficient $\neq$ fair: WTP includes ability to pay.**



Summary

- Use module 1 logic. Surplus is a measure of net benefits
- Prices facilitate trades and reallocation based on Week 1 thinking
- Once the trade happens, price is a way to allocate surplus from that trade between the participants.
- Quantity is primary—always first determine which trades take place given information and then identify all surplus types

Individuals to markets

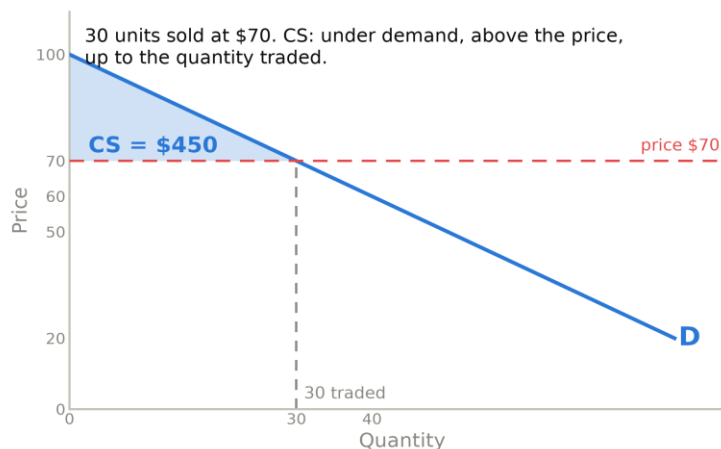
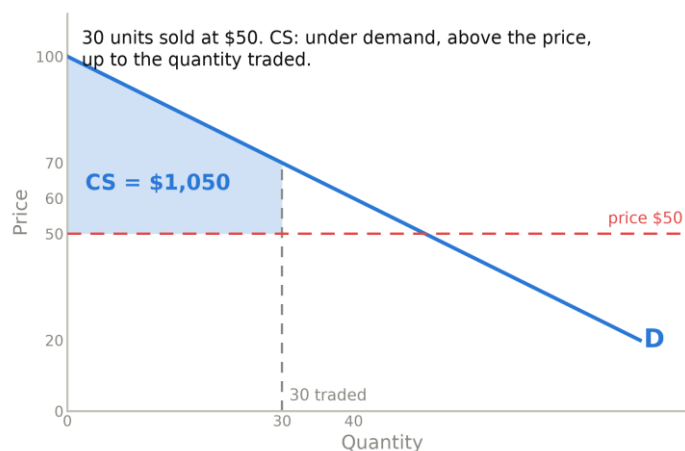
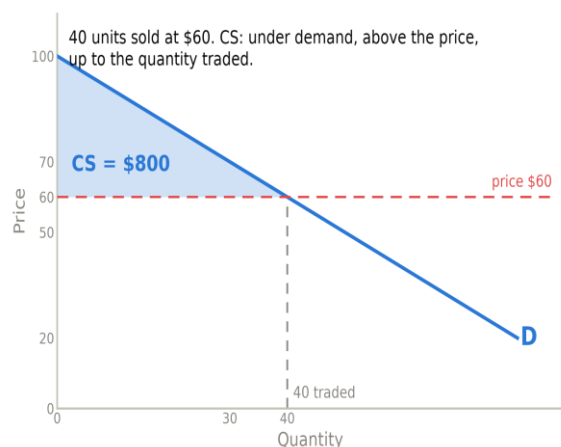
W1+W3's markets as aggregating
individuals $\Rightarrow$ surplus as areas

W3: Individuals to markets

- W3: market demand is a queue of buyers sorted by MWTP. Market supply is a queue of sellers sorted by the least they will accept.
- W1: Total Benefit(Q) as areas under the MWTP curve up to quantity Q .
- Market Surplus is the sum of individual surplus

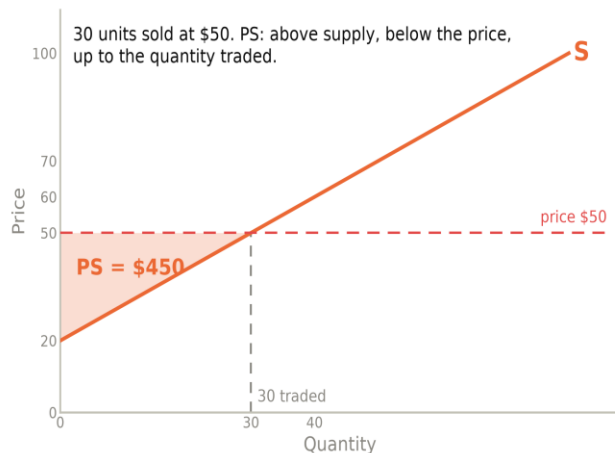
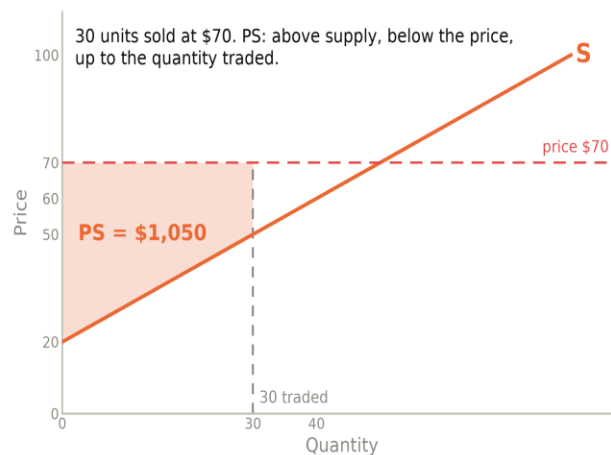
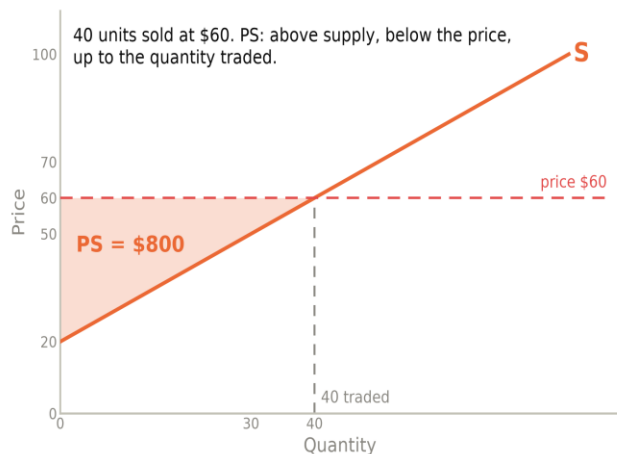
Surplus as areas: CS

- **CS:** between the demand curve and the price, up to the quantity traded.



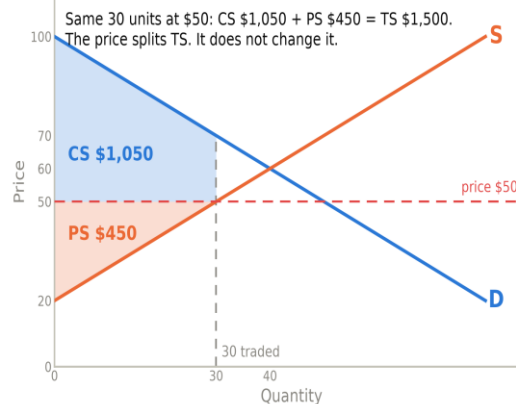
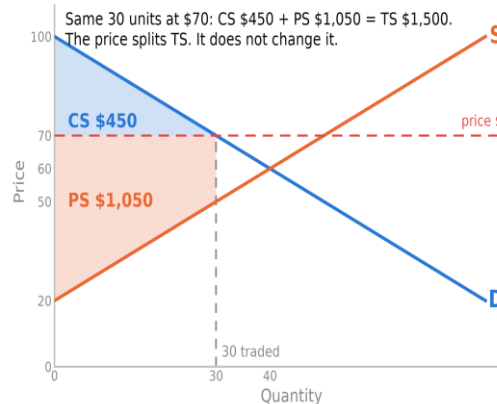
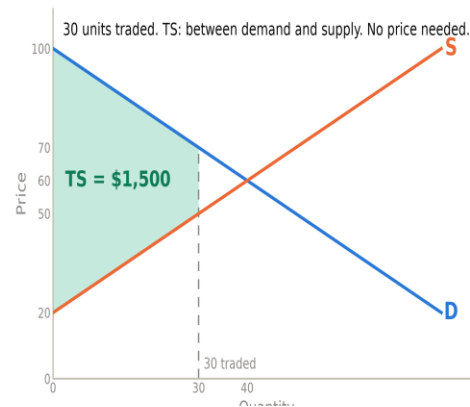
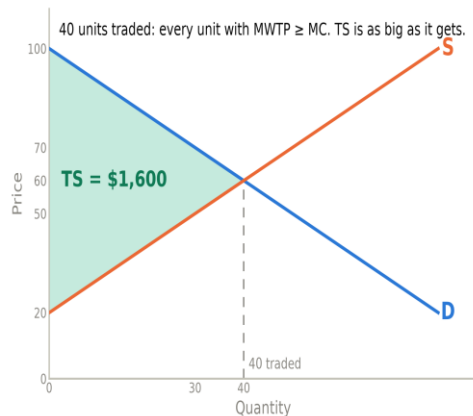
Surplus as areas: PS

- **PS:** between the price and the supply curve, up to the quantity traded.



Surplus as areas: TS

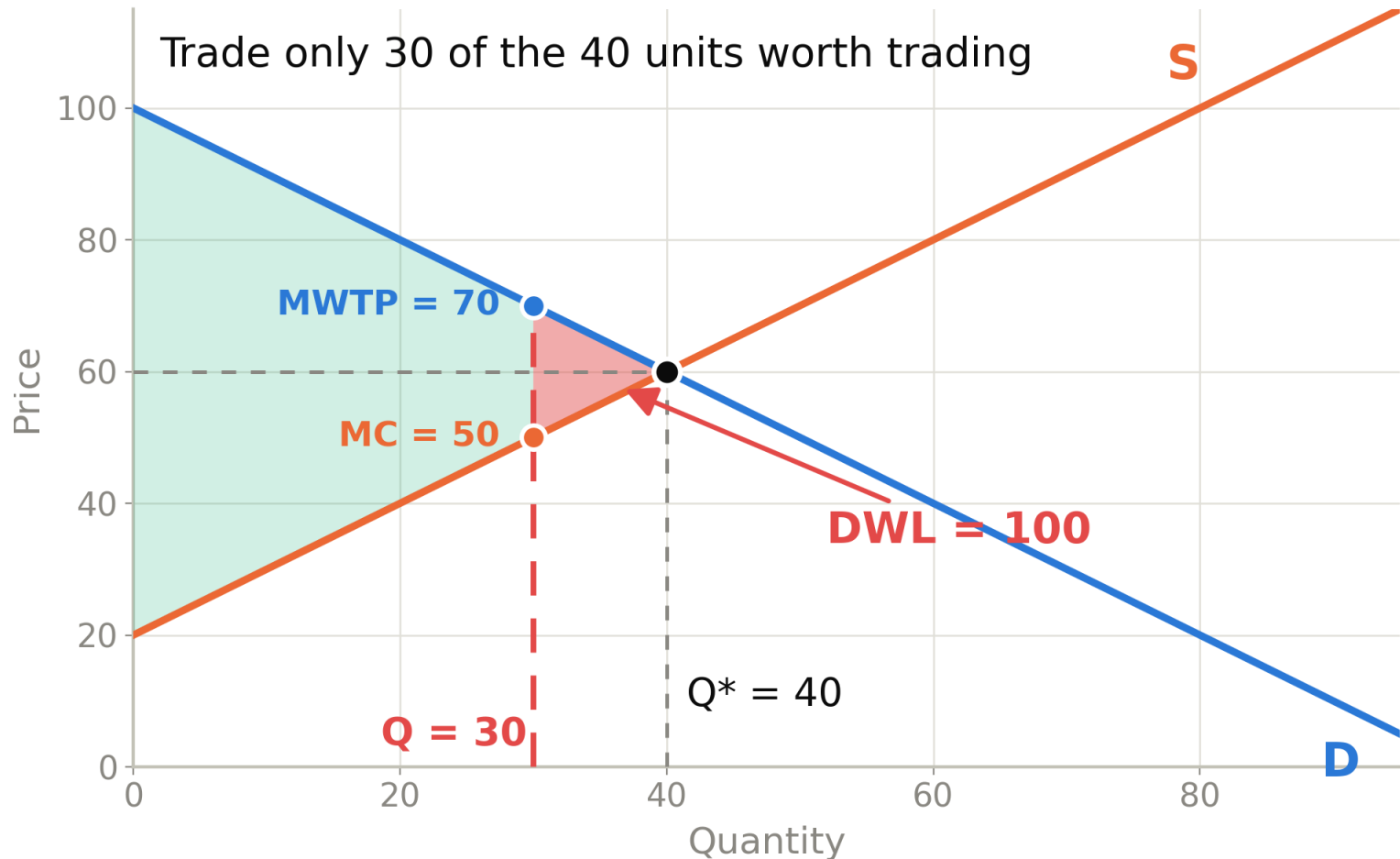
- **TS:** between the Marg. Social Benefit and Marg. Social Cost for the units traded.
 - $MWTP = MSB$ and $MC = MSC$ if no externalities. The price only splits TS into CS and PS.



Why Would TS be less than Efficient?

- Assuming no externalities
- Not producing all and only units for which $MSB \geq MSC$
- Misallocation: Wrong people buying/producing the units traded

Example: Q less than efficient



Misallocation: Why the Demand and Supply diagram can't see it

- W3: market demand is a queue of buyers, sorted from highest MWTP to lowest.
- The area under demand up to Q assumes the Q units went to the first Q people in that queue.
- Lottery: $Q = 3$. On the Demand (and Supply) diagram since everyone is already in order, we can't see if the allocation differs.
- The loss only shows when you look at who holds the units: a gap in the queue. F2 has no seat, F5 has one.

Markets are Efficient

“The market outcome is efficient” true if:

- Everyone is a price taker, and prices are free to move
- Buyers and sellers have full information
- Nobody outside the trade is affected by it
- Trade is voluntary, and no one is excluded by anything but price

We need to check if markets are efficient, not assume they always are

Summary

$|DWL| = \text{Efficient Total Surplus} - \text{Allocation surplus}$
Deadweight Loss (DWL) arises from two sources

1. Over/under production

- Units with a net gain ≥ 0 to **society** not produced
- Units with a net gain < 0 to **society** produced
- Notice only quantity important, not price

2. Misallocation: Keeping quantity fixed, reallocating the good either across buyers or sellers (or both) increases **total surplus**

- Typically, there is an incentive to trade in such cases if allowed/possible

Who gets the surplus when things change?

Can we do better? Efficiency vs. fairness

Dynamic Pricing: FIFA prices tickets so that demand = supply

- Forget the lottery. FIFA sets a price of \$400 and sells to whoever will pay it. Same five fans, same three seats, marginal cost still zero.
- Who buys — and does anyone need to resell?

Same seats, same TS, different split

- At \$400, F1, F2 and F3 buy. Nobody needs to resell: the seats already sit with the three highest values.
- Compare the same three fans getting the seats at the \$100 face value with the best-case allocation

	Price \$100	Price \$400
Consumer surplus	\$1,800	\$900
Producer surplus (FIFA)	\$300	\$1,200
Total surplus	\$2,100	\$2,100

\$900 moves from fans to FIFA. The price splits TS; it does not change it.

Worksheet Q2 (counts for grades)

Put answer on worksheet

Summary

- Quantity first: which trades happen, and who holds the units, decide TS. The price only splits TS into CS and PS.
- Resale fixes misallocation: TS rises and no one in the trade is worse off.
- Pricing high from the start avoids the misallocation but moves surplus from buyers to sellers.

Something to think about at home

Don't forget to give me your
completed worksheets

See you next workshop!
